**Physical Cume**
**December 6, 2002**

*1.)* In group theory, what is a *matrix representation* of a point group, what is the *character* of a representation, how are *classes of conjugate elements* defined, what are *equivalent representations*, and what are *reducible and irreducible representations*?

*2.)* Consider pointgroup $C_{2v}$. Assume a Cartesian coordinate system with x,y,z axes, in which the z-axis is along the $C_2$ axis, and the (z,x)-plane is the paper plane.

*a)* Determine to which irreducible representations of point group $C_{2v}$ the translations along the x-, y-, and z-axes belong.

*b)* If a mass $m$ moves with momentum $\mathbf{p} = m\mathbf{v}$ relative to a fixed point O, then the angular momentum $\mathbf{L}$ of the mass with respect to the point O is the vector product $\mathbf{L} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times m\mathbf{v}$, where $\mathbf{r}$ is the radius vector from O to the position of the particle. The components of $\mathbf{L}$ in the Cartesian coordinate system are defined as follows:

$$L_x = y\,p_z - z\,p_y$$
$$L_y = z\,p_x - x\,p_z$$
$$L_z = x\,p_y - y\,p_x$$

Determine to which irreducible representations of point group $C_{2v}$ the components $L_x$, $L_y$, and $L_z$ belong.

| $\mathscr{C}_{2v}$ | $E$ | $C_2$ | $\sigma_v(xz)$ | $\sigma_v(yz)$ |
|---|---|---|---|---|
| $A_1$ | 1 | 1 | 1 | 1 |
| $A_2$ | 1 | 1 | −1 | −1 |
| $B_1$ | 1 | −1 | 1 | −1 |
| $B_2$ | 1 | −1 | −1 | 1 |